

Phase 8: Data Management & Deployment

Executive Summary

Phase 8 focused on implementing strong **data management** and **deployment practices** for the Smart Service Appointment & Warranty Automation project. The objective was to maintain data quality, prevent duplicates, establish backup processes, and enable smooth migration of metadata from the development org to production.

Key outcomes include:

- Setting up **duplicate rules** for appointments and warranties.
- Scheduling **data export backups**.
- Using both **declarative (Change Sets)** and **developer tools (VS Code & SFDX)** for deployment.
- Understanding package options for future distribution.

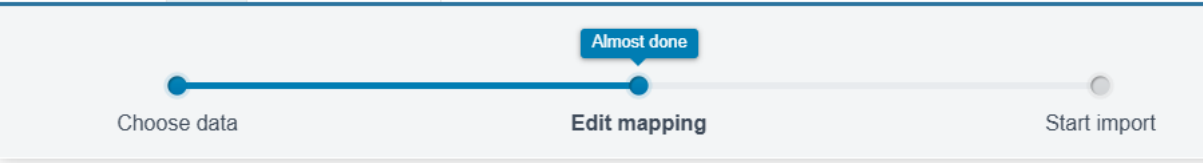
This ensures the application is **reliable, scalable, and production-ready**.

Data Import Wizard

Purpose/Rationale: Browser-based tool for small, quick data loads.

Implementation:

- Used to upload initial sample data for **Technicians, Service Appointments, and Warranties**.
- CSV files with customer details, appointment dates, technicians, warranty IDs, and product serial numbers were imported.
- Field mapping ensured accuracy, and error logs were checked after upload.



Edit Field Mapping: Service Appointments

Your file has been auto-mapped to existing Salesforce fields, but you can edit the mappings if you wish. Unmapped fields will not be imported. [Help for this page](#)

Edit	Mapped Salesforce Object	CSV Header	Example	Example	Example
Change	Record Owner	Customer_Name__c	John Doe	Jane Smith	Michael Lee
Change	Record Type	Appointment_Date__c	2025-09-15	2025-09-20	2025-09-27
Change	Service Appointmen...	Technician__c	Tech-001	Tech-002	Tech-003
Change	Service Appointmen...	Service_Type__c	Installation	Repair	Maintenance
Change	Technician	Status__c	Scheduled	Completed	Scheduled
Change	Record Owner	Warranty_Id__c	W-1001	W-1002	W-1003
Change	Service Appointmen...	Product__c	AC Unit	Washing Machine	Refrigerator
Change	Service Appointmen...	Serial_Number__c	SN-AC1234	SN-WM4567	SN-RF7890

CancelPreviousNext

SetupHomeObject Manager

Great job

Choose dataEdit mappingStart import

Review & Start Import

Review your import information and click Start Import.

Your selections:

Service Appointments ✓
Add new records ✓
sample_service_warranty_data.csv ✓

Your import will include:

Mapped fields8

Your import will not include:

Unmapped fields0

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Bulk Data Load Jobs

SETUP

Bulk Data Load Job

75dgK00000De06g

[View the details of a bulk data load job.](#)

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Bulk Data Load Job Detail [Refresh](#)

Job ID	75dgK00000De06g	Job Type	Bulk V1	Status	Closed
Submitted By	Ethaatha Solla	Operation	Uppert	Total Processing Time (m:s)	65
Start Time	9/23/2025, 8:47 AM PST	Queued Batches	0	API Active Processing Time (m:s)	0
End Time	9/23/2025, 8:47 AM PST	In Progress Batches	0	Apex Processing Time (m:s)	0
Time to Complete (hh:mm:ss)	00:01	Completed Batches	1		
Object	Service Appointment	Failed Batches	0		
External ID Field	Name	Progress	100%		
Contact Type	CSV	Records Processed	5		
Concurrency Mode	Parallel	Records Failed	5		
API Version	64.0	Retries	0		

[Refresh](#)

Batches

View Request	View Result	Batch ID	Start Time	End Time	Total Processing Time (m:s)	API Active Processing Time (m:s)	Apex Processing Time (m:s)	Records Processed	Records Failed	Retry Count	State Message	Status
View Request	View Result	75dgK00000B3LET	9/23/2025, 8:47 AM	9/23/2025, 8:47 AM	65	0	0	5	5	0		Completed

Duplicate Rules

Purpose/Rationale: Maintain clean data by preventing duplicate appointments and warranty entries.

Implementation:

- **Service Appointments:** Created a Matching Rule on **Customer__c + Appointment_Date__c**.
- **Warranties:** Created a Matching Rule on **Serial_Number__c + Product__c**.
- Duplicate Rules configured with action **Allow, but Alert and Report**, so users are warned while admins can review duplicate sets.

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Q matching

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▼ Duplicate Management

Matching Rules

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Matching Rules

Matching Rule

Service_Appointment_Match_Rule

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Matching Rule Detail

EditDeleteCloneActivate

Object	Service Appointment		
Rule Name	Service_Appointment_Match_Rule		
Unique Name	Service_Appointment_Match_Rule		
Description			
Matching Criteria	(Service Appointment: Contact EXACT MatchBlank = FALSE) AND (Service Appointment: Name EXACT MatchBlank = FALSE)		
Status	Inactive		
Created By	Bhavitha Bolla, 9/23/2025, 9:09 AM	Modified By	Bhavitha Bolla, 9/23/2025, 9:09 AM

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Matching Rules

Matching Rule

Warranty_Match_Rule

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Matching Rule Detail

DeleteCloneDeactivate

Object	Warranty		
Rule Name	Warranty_Match_Rule		
Unique Name	Warranty_Match_Rule		
Description			
Matching Criteria	(Warranty: Contact EXACT MatchBlank = FALSE) AND (Warranty: Name EXACT MatchBlank = FALSE)		
Status	Active		
Created By	Bhavitha Bolla, 9/23/2025, 9:11 AM	Modified By	Bhavitha Bolla, 9/23/2025, 9:11 AM

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Duplicate Error Logs

Duplicate Rules

Matching Rules

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Duplicate Rules

Service Appointment Duplicate Rule

Service_Appointment_Duplicate_Check

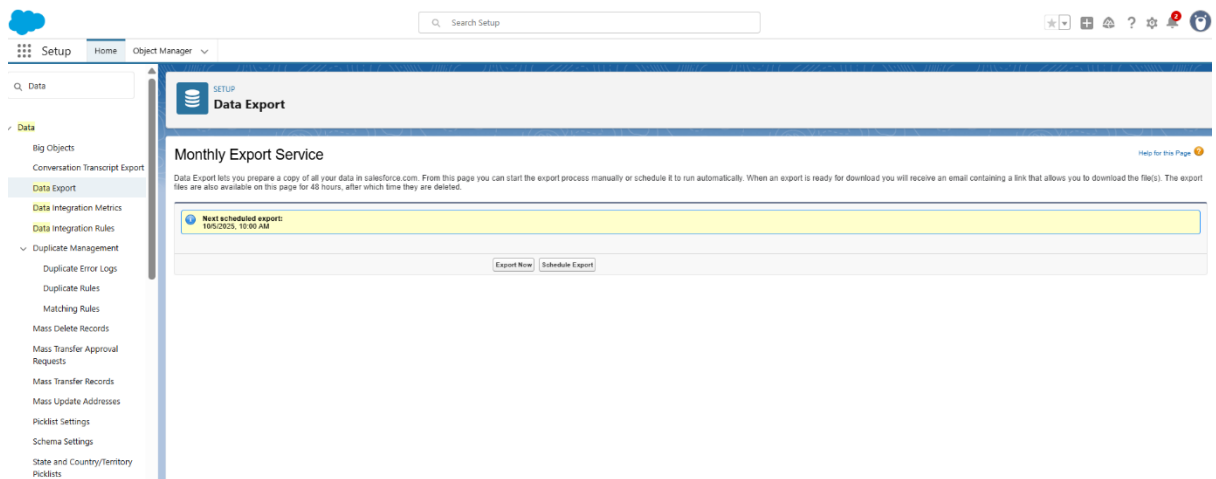
Help for this Page

Duplicate Rule Detail

EditDeleteCloneActivate

Rule Name	Service_Appointment_Duplicate_Check	Order	1 of 1 [Reorder]	i
Description				
Object	Service Appointment			
Record-Level Security	Enforce sharing rules			
Action On Create	Allow	Operations On Create	☒ Alert ☒ Report	
Action On Edit	Allow	Operations On Edit	☐ Alert ☐ Report	
Alert Text	"Possible duplicate appointment detected – same customer and date."			
Active	☐			
Matching Rule	✔ Service Appointment Match Rule ✔ Mapped	Matching Criteria	(Service Appointment: Contact EXACT MatchBlank = FALSE) AND (Service Appointment: Name EXACT MatchBlank = FALSE)	
Conditions				
Created By	Bhavitha Bolla, 9/23/2025, 9:14 AM		Modified By	Bhavitha Bolla, 9/23/2025, 9:14 AM

EditDeleteCloneActivate



Change Sets

Purpose/Rationale: Move metadata between sandbox and production.

Implementation:

- Outbound Change Sets used to migrate **Apex triggers, Lightning Web Components, and Permission Sets**.
- Imported and validated in Production before deployment.

Unmanaged vs Managed Packages

Purpose/Rationale: Future packaging for sharing the solution.

Implementation:

- **Unmanaged Package:** For internal org-to-org movement during testing.
- **Managed Package:** Considered for distributing the Smart Service Automation solution to external businesses, ensuring IP protection and upgrade path.

ANT Migration Tool

Purpose/Rationale: Command-line deployment option.

Implementation:

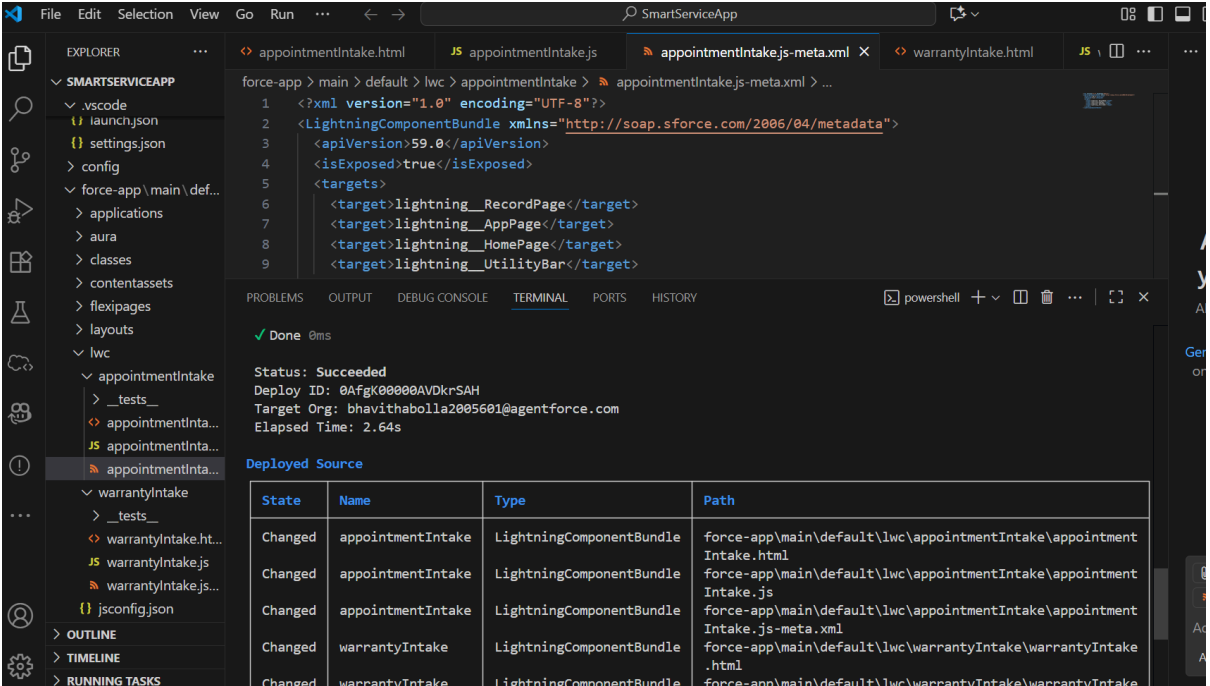
- Evaluated ANT for **scriptable deployments** via package.xml and build.xml.
 - While powerful, chosen less frequently compared to VS Code & SFDX for this project.
-

VS Code & SFDX

Purpose/Rationale: Modern source-driven development & deployment.

Implementation:

- Developed Apex classes, triggers, and LWCs in **VS Code** with Salesforce Extensions.
- Version control managed in **Git**.
- Deployment executed using:
- `sfdx force:source:deploy -p force-app`
- Ensured seamless deployment and alignment with **CI/CD practices**.



force-app > main > default > lwc > appointmentIntake > appointmentIntake.js-meta.xml > ...

```
1 <?xml version="1.0" encoding="UTF-8"?>
2 <LightningComponentBundle xmlns="http://soap.sforce.com/2006/04/metadata">
3   <apiVersion>59.0</apiVersion>
4   <isExposed>true</isExposed>
5   <targets>
6     <target>lightning__RecordPage</target>
7     <target>lightning__AppPage</target>
8     <target>lightning__HomePage</target>
9     <target>lightning__UtilityBar</target>
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS HISTORY

✓ Done 0ms

Status: Succeeded
Deploy ID: 0AfgK00000AVDkrSAH
Target Org: bhavithabolla2005601@agentforce.com
Elapsed Time: 2.64s

Deployed Source

State	Name	Type	Path
Changed	appointmentIntake	LightningComponentBundle	force-app\main\default\lwc\appointmentIntake\appointmentIntake.html
Changed	appointmentIntake	LightningComponentBundle	force-app\main\default\lwc\appointmentIntake\appointmentIntake.js
Changed	appointmentIntake	LightningComponentBundle	force-app\main\default\lwc\appointmentIntake\appointmentIntake.js-meta.xml
Changed	warrantyIntake	LightningComponentBundle	force-app\main\default\lwc\warrantyIntake\warrantyIntake.html
Changed	warrantyIntake	LightningComponentBundle	force-app\main\default\lwc\warrantyIntake\warrantyIntake.js